



Seqela

BANKING TECHNOLOGY · ETHIOPIA

TECHNICAL SPECIFICATION

PREPARED FOR

DECSI

Dedebit Credit and Savings Institution S.C. · Mekelle, Tigray

Technical Specification AI-powered Credit Intelligence

Functional and technical specification of the DECSI credit factory — architecture, modules, integrations, security, deployment, and acceptance criteria.

Django 3.2 · Python 3.9

PostgreSQL 16

Docker

CBS customer API

OCR eng+amh

Offline field PWA

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Document control

ITEM	DETAIL
Audience	DECSI IT, Credit Operations, Collateral/Engineering, Risk & Compliance, Executive sponsors
Purpose	Define what the system is, how it works, how it deploys, and how DECSI accepts it
Baseline	Running production-shaped codebase designed around DECSI Branch → District → HO practices
Related docs	Business proposal <i>DECSI_AI_Credit_Intelligence_Proposal.pdf</i> ; live UI screenshots

1. Purpose, Scope & Document Control

Purpose: Specify the AI-powered Credit Intelligence platform for DECSI — a digital credit factory for loan intake, document authentication, 7-step cashflow appraisal (MSME/corporate), GPS collateral valuation, multi-level committees, and post-approval disbursement tracking.

1.1 In scope

- Staff web application (branch / district / HO roles)
- Loan request lifecycle and role queues
- Document checklist with OCR (English + Amharic) and authentication workflow
- Appraisal Sheets 1–7 with Excel-faithful completeness gates, scorecard, analysis assist
- Collateral catalog, field visit (online + offline PWA), coverage, lock/unlock, evidence pack
- Configurable Branch → District → Head Office → Management committee voting
- CBS customer number lookup / smart prefill (and optional transactions feed)
- Docker-based deployment on DECSI infrastructure (on-prem or private cloud)

1.2 Out of scope (unless separately contracted)

- Full Temenos/CBS rip-and-replace or account booking engine
- Customer-facing retail mobile banking app
- General ledger / IFRS reporting suite
- Hardware (tablets, GPS devices) procurement

Design principle: The Hub is DECSI's *credit factory*. Core banking remains the system of record for accounts and fund disbursement.

2. System Overview & Lifecycle

End-to-end secured / cashflow credit processing for MSME and corporate facilities:



STAGE	PRIMARY ACTORS	SYSTEM BEHAVIOUR
Intake	Branch manager / officer	Create LoanRequest ; product, amount, collateral type, customer #
Documents	Officer / engineer	Required types; OCR; identity match; verify/reject
Pre-committee queues	Ops / finance managers	Optional approval flags before appraisal assignment
Appraisal	Loan officer	Guided Sheets 1-7; MSME or Corporate mode; policy gates
Collateral	Officer and/or engineer	Building BOQ / land / movable; GPS photos; offline sync
Committee	Voters by level	Pack, vote, return-to-officer; amount routing
Post-approval	Ops / finance / officer	Conditions → schedule → ready → disbursed (CBS booking external)

3. Logical Architecture



Application packages

- `decsi_loan/` — settings, URLs, WSGI
- `loans/` — credit factory core
- `collateral/` — valuation & field PWA
- `templates/` , `static/` — UI & PWA assets

Runtime topology

- `web` — Django app container
- `db` — PostgreSQL 16
- Optional `nginx` HTTPS overlay for tablet secure context (GPS/camera)

4. Technology Stack

LAYER	TECHNOLOGY	NOTES
Language	Python 3.9	Container base image
Framework	Django 3.2.x	Server-rendered staff UI
Database	PostgreSQL 16	psycopg2-binary
Auth	Django sessions + loans.CustomUser	Role field RBAC
Frontend	Django templates, jQuery 3.6, CSS	No SPA framework required
OCR	Tesseract + pytesseract + pdf2image	Default eng+amh
PDF	WeasyPrint, ReportLab	Committee / evidence packs
Excel I/O	pandas, openpyxl, django-import-export	Catalog & bulk masters
HTTP	requests	CBS party / transactions
Maps	Gebeta Maps + MapLibre; Leaflet/OSM fallback	Field GPS & geocoding
Offline	Service Worker + IndexedDB PWA	Collateral field capture
Deploy	Docker Compose	web + db (+ optional nginx TLS)

DRF / JWT packages may appear in requirements.txt for future API work; current staff UI is session-based Django views.

5. Functional Modules (Detailed)

MODULE	CAPABILITIES
A. Intake & master data	Loan requests; categories/products; collateral types; geography (Region → Zone → City → District → Branch); user administration; CSV bulk import commands
B. Documents	Configurable required types (MSME/corporate aware); size/MIME checks; OCR eng+amh; identity match (name/phone/TIN); statuses Pending → Auto-passed / Needs review → Verified / Rejected; replace-aware upload
C. Appraisal	7-step wizard; MSME vs Corporate mode; qualitative ≥75% gate; DSCR/stress/capacity; E&S 21 items; scorecard /100; analysis assist; amortization schedule; finish gates
D. Collateral	Building BOQ catalog × woreda unit prices; land; movable; GPS site mark; photo rules; EXIF vs live GPS; lock after submit; supervised unlock; engineering QA; coverage vs ask; evidence pack (Print/PDF/JSON/ZIP)
E. Approvals	Ops/finance queues; amount-routed committee levels; vote Approve/Decline; return-to-officer; notifications; read-only committee pack
F. Post-approval	Conditions clearance → schedule readiness → ready to disburse → disbursed flag (CBS booking external)
G. Reporting	Management report generation options by role

5.1 Appraisal mode

Mode is driven by `LoanCategory.appraisal_mode ( msme / corporate )` and stamped on `LoanAppraisal`. Corporate mode changes Sheet 1 KYC depth, Sheet 2 qualitative factor set (governance), and Sheet 3 financial statement emphasis while keeping the same 7-step wizard and committee factory.

6. Data Domains & Key Entities

Credit core

- `LoanRequest` — central case file
- `LoanAppraisal` — 1:1 appraisal record
- `LoanRequestBasicInfo` — Sheet 1 + field provenance
- Qualitative factors, E&S items, amortization lines
- Documents & document types / policies
- Committee progress, votes, notifications

Collateral & org

- Building / Land / Other collateral + images
- `MainWork` → `SubWork` → `SubSubWork` + unit prices
- `CollateralPolicyConfig`, audit log, unlock requests
- `CustomUser` + roles; District / Branch masters
- `LoanCategory`, `CollateralType`

6.1 Provenance

Sheet 1 fields retain source badges: **Core banking / OCR / Document / Officer** / defaults. CBS conflicts are listed for Accept (banking wins) rather than silent overwrite.

7. Appraisal Engine (Sheets 1–7)

SHEET	MSME FOCUS	CORPORATE FOCUS	HARD GATES (TYPICAL)
1	Basic info & loan request	Entity KYC, directors, UBO	Required KYC completeness
2	Business & character (10 factors)	Governance & character	Qualitative $\geq 75\%$
3	Cashflow / DSCR / stress / 12-mo grid	FS + cashflow + ratios	Annual / stressed DSCR < 1.0 block
4	E&S — 21 items, 5 sections		E&S REJECT hard block
5	Collateral worksheet (from field valuation)		Coverage warning at policy min
6	Summary, 4-pillar scorecard, recommendation		Incomplete sheets 1–6 cannot finish
7	Repayment & amortization		Schedule for booking / pack

7.1 Scorecard pillars (explainable /100)

PILLAR	WEIGHT	INPUTS
Qualitative / character	25	Sheet 2 totals
Financial / cashflow	35	DSCR, stress DSCR, capacity vs ask, ratios
Environmental & social	15	Risk category & eligibility
Collateral coverage	25	Coverage vs ask (e.g. $\geq 1.5\times$ full points)

Bands: Strong ≥ 80 · Acceptable ≥ 60 · Weak ≥ 40 · Unacceptable < 40 . **Analysis Assist** surfaces thin pillars and hard blocks — decision support, not auto-approve.

Key modules: `loans/appraisal_mode.py`, `cashflow_utils.py`, `sheet_requirements.py`, `appraisal_scorecard.py`, `analysis_assist.py`.

8. Collateral, GPS & Offline PWA

8.1 Field visit

- Wizard: Site → BOQ / valuation → Photos → Review
- GPS accuracy capture; map pin (Gebeta or Leaflet/OSM)
- Default policy examples: min 5 building / 3 land photos; weak accuracy threshold ~100 m; photo max distance from site ~200 m; declared-address mismatch flags
- EXIF GPS vs live capture mismatch warnings; optional hard blocks
- Submit locks dossier for audit; supervised unlock workflow

8.2 Offline Progressive Web App

COMPONENT	BEHAVIOUR
Manifest / SW	<code>manifest-collateral.webmanifest</code> , <code>collateral_sw.js</code>
Local store	IndexedDB bundle for site data & photos
Sync	Prepare offline → capture → Sync now when online
Secure context	Optional HTTPS nginx overlay for camera/GPS on LAN tablets

DECSI fit: Offline tablet path is required for rural branches with intermittent connectivity — core to DECSI's field reality.

9. Committee & Disbursement Track

9.1 Committee engine

- Configurable levels: Branch → District → Head Office → Management
- Amount used for routing: Sheet 6 recommended amount (fallback: requested)
- Member rules: by role scoped to branch/district/org, or named users
- Thresholds: min approvals / declines per level (configurable)
- Outcomes: Approve · Decline · Return to officer with notes
- Read-only appraisal pack for voters (Sheets 1–7 summary)

9.2 Disbursement track (in-app)

Post-final-approval states track conditions clearance, schedule readiness, ready-to-disburse, and disbursed.

Fund release and account booking remain in CBS via operational handoff — the Hub does not replace the ledger.

10. Integrations

10.1 DECSI CBS — customer party lookup

ITEM	SPECIFICATION
Config	DECSI_BASE_URL , timeout, mock/fallback flags
Typical path	/getCusByCusNo/api/v1.0.0/party/custid/{cid}/custdets
Use	Prefill Sheet 1 empty fields; conflict list for Accept
Service	loans/services/customer.py

10.2 Transactions (optional banking pillar)

Optional path via DECSI_TRANSACTIONS_PATH for banking behaviour signals in scoring/assist (loans/services/banking_transactions.py).

10.3 OCR & maps

- OCR language: DOCUMENT_OCR_LANG (default eng+amh)
- Gebeta Maps API for geocode/tiles; Nominatim / Leaflet+OSM fallbacks

Interface ownership: DECSI IT provides CBS sandbox credentials, network allow-lists, and field mapping for customer attributes. Seqela implements adapters and conflict UX.

11. Roles, Access Control & Security

11.1 Role catalogue (out of the box)

FAMILY	ROLES
Admin	superadmin, admin
Branch credit	loan_officer, branch_manager, accountant
Engineering	engineer, engineering_head
Pre-committee	operation_manager, finance_manager
District / HO	district_manager, credit_committee
Executive	ceo, vp, board_member
Control	risk_compliance, auditor

Users may be scoped to `branch` and/or `district` . Committee eligibility follows level rules.

11.2 Security controls

- Authenticated session access on all operational screens
- Role checks on views / workflows
- Document authentication trail and field provenance
- Collateral lock + supervised unlock + field audit log
- Hard/soft policy gates before finish / committee
- Evidence pack / dossier export for audit reconstruction
- Bank-controlled data residency (on-prem / private cloud Docker)

12. Non-Functional Requirements

AREA	TARGET / REQUIREMENT
Availability	Business-hours critical; overnight batch windows agreed with DECSI IT
Performance	Interactive pages <3s typical on LAN; OCR may be async/longer for large PDFs
Concurrency	Sized for DECSI concurrent branch users (capacity planning in discovery)
Offline	Field PWA must capture GPS/photos offline and sync without data loss
Backup	PostgreSQL backup/restore procedures owned with DECSI IT
Auditability	Who/when for documents, appraisal finish, locks, unlocks, votes
Localisation	UI English; OCR supports Amharic; bilingual labels extendable
Browser	Current Chrome/Edge/Firefox; Chromium-based for best PWA field support

13. Deployment, Environments & Ops

13.1 Standard topology

SERVICE	IMAGE / ROLE	PORT
web	App build from Dockerfile (Python 3.9 + Tesseract eng/amh + WeasyPrint libs)	8000
db	postgres:16 with persistent volume	5432 (internal)
nginx (optional)	TLS terminator for tablet HTTPS (docker-compose.https.yml)	8443 → 443

13.2 Environments (recommended)

- **Dev / UAT** — configuration & training; mock or sandbox CBS
- **Production** — DECSI-controlled host; real CBS; backup & monitoring

13.3 Configuration (non-secret names)

SECRET_KEY , DEBUG , SITE_URL , DB_* , DECSI_BASE_URL , DECSI_CUSTOMER_TIMEOUT , DECSI_CUSTOMER_*_MOCK , DECSI_TRANSACTIONS_PATH , DOCUMENT_OCR_LANG , GEBETA_MAPS_* , USE_HTTPS_PROXY , CSRF_TRUSTED_ORIGINS , email settings.

13.4 Operations

- Migrations via Django manage.py
- Bulk master import management commands (zones, branches, users, catalogs)
- Application logs + Postgres backups; hypercare during network rollout

14. Configuration & Bank Policy Knobs

DECSI Credit/Risk can tune without code changes (via admin / policy models):

Credit / analysis

- Loan categories & appraisal mode
- Document type requirements
- Qualitative pass threshold
- DSCR / stress hard blocks
- Scorecard weights (if enabled)
- Committee levels, voters, thresholds

Collateral / field

- Min photos by asset type
- GPS accuracy / distance rules
- Coverage minimum & flag ratio
- Block-on-submit toggles
- Unit-price catalog by woreda
- Who estimates (officer / eng / both)

15. Assumptions, Exclusions & Interfaces

15.1 Assumptions

- DECSI provides infrastructure (VM/hosts), TLS certificates if required, and CBS connectivity
- Cashflow MSME methodology remains aligned to DECSI practice and the sector Excel standard
- Officers and engineers receive role-based training before go-live of each wave

15.2 Interface responsibilities

PARTY	RESPONSIBILITY
DECSI IT	Hosting, network, backups, CBS API access, AD/SSO roadmap (if later required)
DECSI Credit	Policy thresholds, products, committee design, UAT sign-off
Seqela	Application delivery, configuration, training, defect fix, methodology updates

15.3 Explicit exclusions

- CBS disbursement posting API (handoff process only, unless separately specified)
- National ID / NBE bureau live feed (flags captured on file; live bureau connector optional future work)

16. Acceptance Criteria & Test Approach

16.1 Functional acceptance (must pass)

1. Create loan → upload required docs → OCR/auth statuses visible
2. CBS customer # prefill populates empty Sheet 1 fields; conflicts Acceptable
3. Complete MSME Sheets 1–7; qualitative <75% and E&S REJECT and DSCR<1.0 block finish as configured
4. Corporate mode switches labels/factors for a corporate category loan
5. Field visit captures GPS + min photos; offline capture syncs when online
6. Coverage ratio appears on Sheet 5 / evidence pack
7. Submit to committee; voter sees pack; Approve/Decline/Return works; routing follows recommended amount
8. Role isolation: auditor/risk read paths; officers cannot cast HO votes without eligibility

16.2 Non-functional acceptance

- UAT environment restored from backup successfully (drill)
- HTTPS field tablet can access camera/GPS (if HTTPS overlay used)
- Production deploy runbook executed with DECSI IT sign-off

16.3 Test types

Unit tests (cashflow helpers, gates, modes) · Integration tests (CBS mock) · UAT scripts by role · Parallel-run sample files vs Excel baseline for credit sign-off.

Next step: Joint TS walkthrough with DECSI IT + Credit → confirm CBS field map → freeze UAT script → schedule environment build.

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